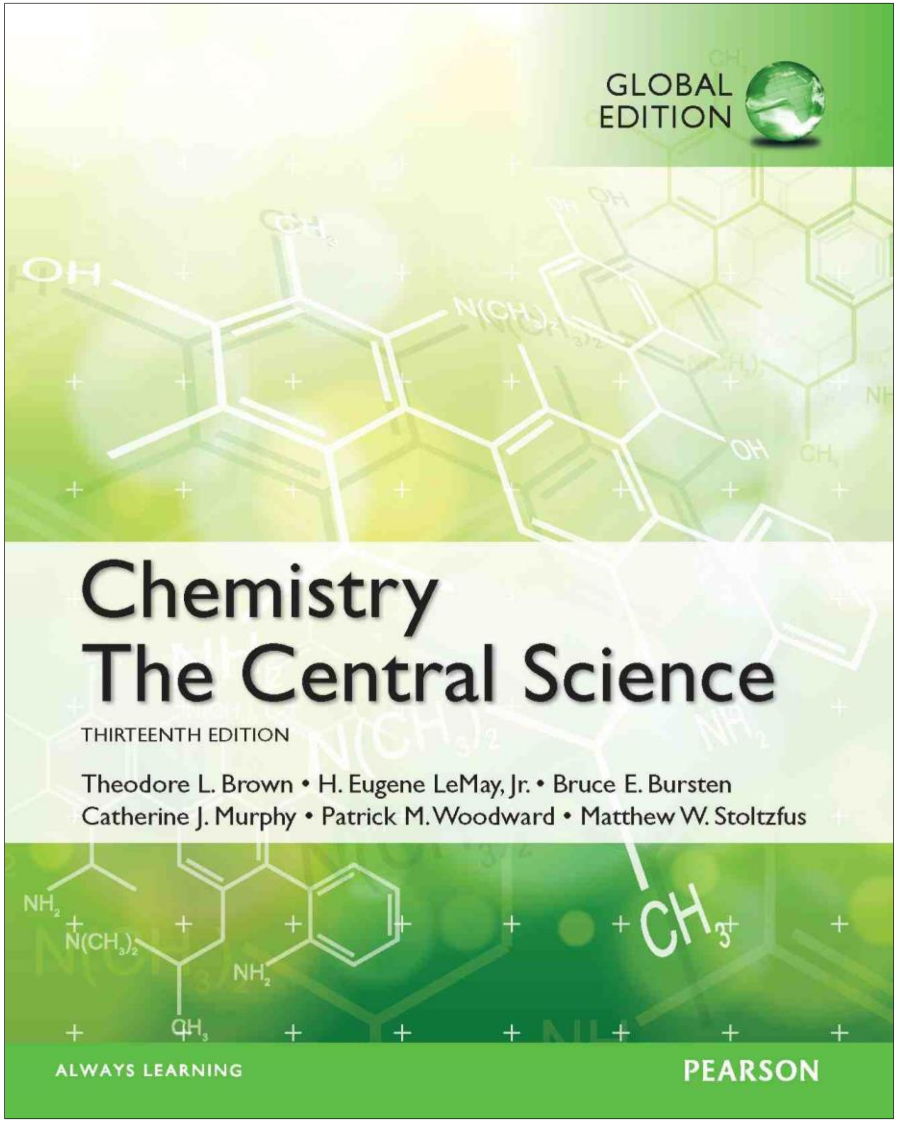


Lecture Presentation

Chapter 16

Acid–Base Equilibria

James F. Kirby
Quinnipiac University
Hamden, CT

Some Definitions

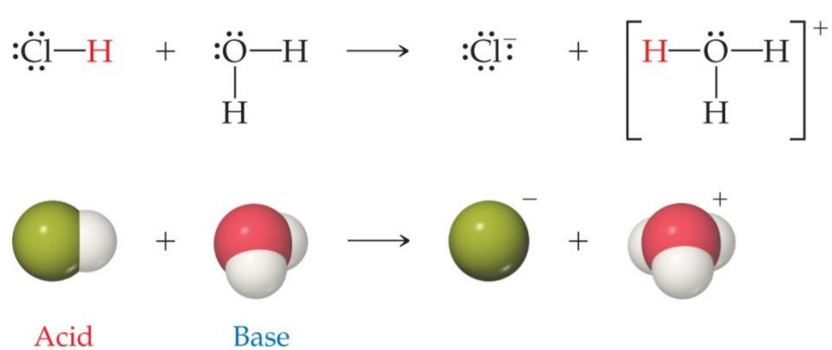
- Arrhenius
  - An acid is a substance that, when dissolved in water, increases the concentration of hydrogen ions.
  - A base is a substance that, when dissolved in water, increases the concentration of hydroxide ions.
- Brønsted–Lowry
  - An acid is a proton donor.
  - A base is a proton acceptor.

Brønsted–Lowry Acid and Base

A Brønsted–Lowry **acid** must have at least one removable (acidic) proton (H^+) to donate.

A Brønsted–Lowry **base** must have at least one nonbonding pair of electrons to accept a proton (H^+).

$$HCl(g) + H_2O(l) \longrightarrow Cl^-(aq) + H_3O^+(aq)$$



What Is Different about Water?

- Water can act as a Brønsted–Lowry base and accept a proton (H^+) from an acid, as on the previous slide.
- It can also donate a proton and act as an acid, as is seen below.
- This makes water **amphiprotic**.

$$\underset{\text{Base}}{NH_3(aq)} + \underset{\text{Acid}}{H_2O(l)} \rightleftharpoons NH_4^+(aq) + OH^-(aq)$$

amphoteric – behaving as an acid and as a base

Al_2O_3 can react with an acid and a base.

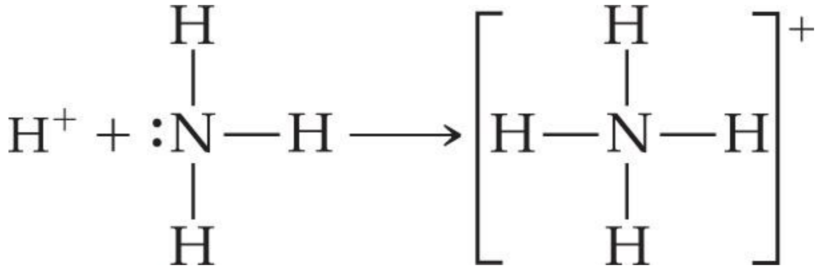
Acids and Bases

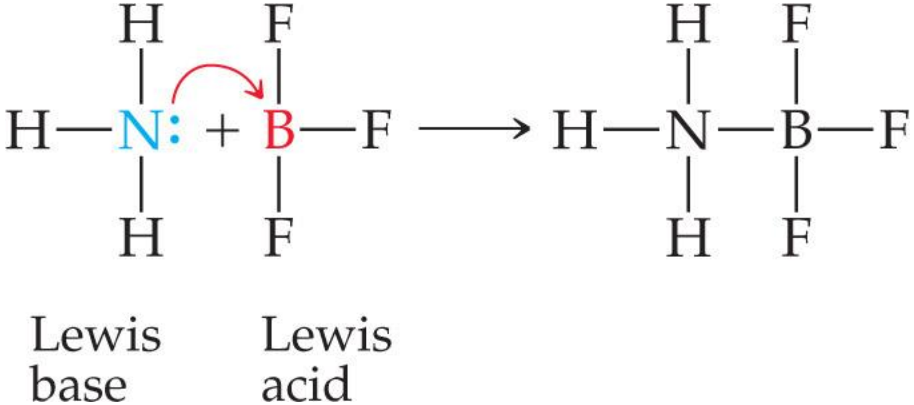
Lewis Acid/Base Chemistry

- Lewis acids are **electron pair acceptors**.
- Lewis bases are **electron pair donors**.

- There are compounds which do *not* meet the Brønsted–Lowry definition which meet the Lewis definition.

Comparing Ammonia's Reaction with H^+ and BF_3



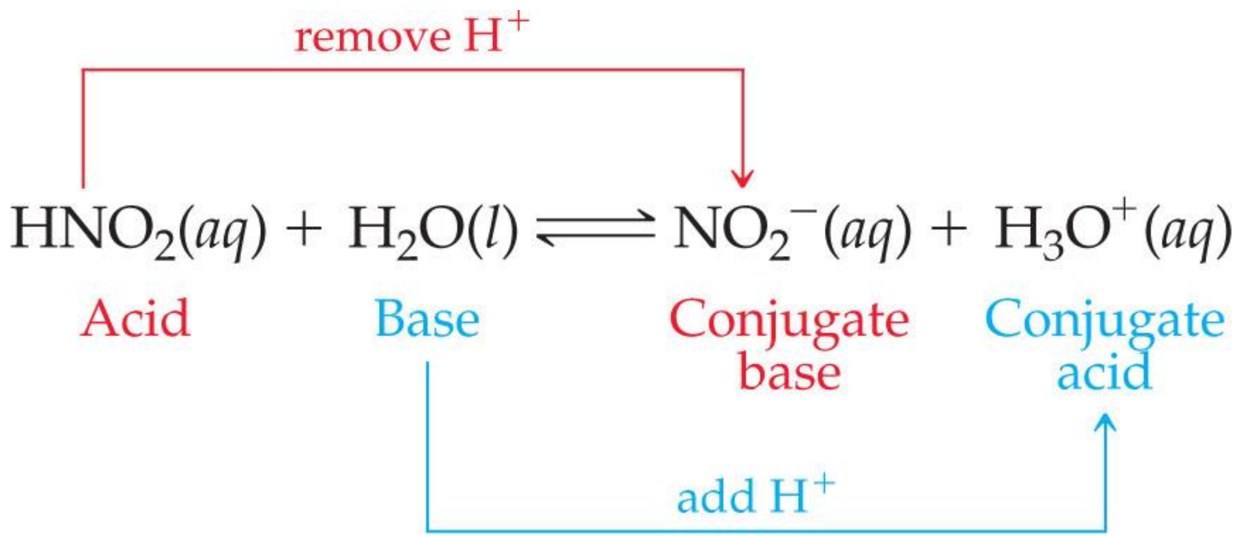


Comparison of Three Acid/Base Definitions

Definition	Acid	Base
Arrhenius	H^+ producer	OH^- producer in water
Bronsted-Lowry	H^+ donor	H^+ accepter
Lewis	e^- pair acceptor	e^- pair donor

Conjugate Acids and Bases

- The term **conjugate** means “joined together as a pair.”
- Reactions between acids and bases always yield their conjugate bases and acids.



SAMPLE EXERCISE 16.1 Identifying Conjugate Acids and Bases

(a) What is the conjugate base of $HClO_4$, H_2S, PH_4^+, HCO_3^-?

(b) What is the conjugate acid of CN^-, SO_4^{2-}, H_2O, HCO_3^-?

SAMPLE EXERCISE 16.1 Identifying Conjugate Acids and Bases

(a) What is the conjugate base of $HClO_4$, H_2S, PH_4^+, HCO_3^-?

(b) What is the conjugate acid of CN^-, SO_4^{2-}, H_2O, HCO_3^-?

Solve

(a) If we remove a proton from $HClO_4$, we obtain ClO_4^-, which is its conjugate base. The other conjugate bases are HS^-, PH_3, and CO_3^{2-}.

(b) If we add a proton to CN^-, we get HCN, its conjugate acid. The other conjugate acids are HSO_4^-, H_3O^+, and H_2CO_3. Notice that the hydrogen carbonate ion (HCO_3^-) is amphiprotic. It can act as either an acid or a base.

SAMPLE EXERCISE 16.2 Writing Equations for Proton-Transfer Reactions

The hydrogen sulfite ion (HSO_3^-) is amphiprotic. Write an equation for the reaction of HSO_3^- with water **(a)** in which the ion acts as an acid and **(b)** in which the ion acts as a base. In both cases identify the conjugate acid–base pairs.

SAMPLE EXERCISE 16.2

Writing Equations for Proton-Transfer Reactions

The hydrogen sulfite ion (HSO_3^-) is amphiprotic. Write an equation for the reaction of HSO_3^- with water **(a)** in which the ion acts as an acid and **(b)** in which the ion acts as a base. In both cases identify the conjugate acid–base pairs.

Solve

(a) $$HSO_3^-(aq) + H_2O(l) \rightleftharpoons SO_3^{2-}(aq) + H_3O^+(aq)$$

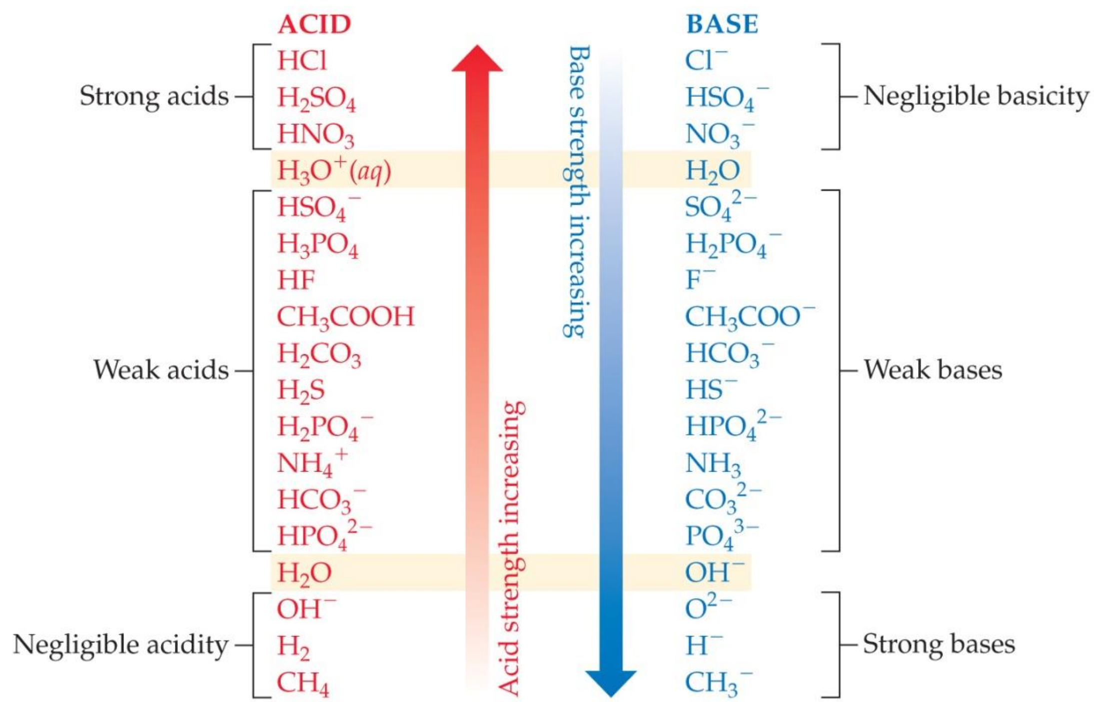
The conjugate pairs in this equation are HSO_3^- (acid) and SO_3^{2-} (conjugate base), and H_2O (base) and H_3O^+ (conjugate acid).

(b) $$HSO_3^-(aq) + H_2O(l) \rightleftharpoons H_2SO_3(aq) + OH^-(aq)$$

The conjugate pairs in this equation are H_2O (acid) and OH^- (conjugate base), and HSO_3^- (base) and H_2SO_3 (conjugate acid).

Relative Strengths of Acids and Bases

- Acids above the line with H_2O as a base are *strong acids*; their conjugate bases do *not* act as acids in water.
- Bases below the line with H_2O as an acid are *strong bases*; their conjugate acids do *not* act as acids in water.



- The substances between the lines with H_2O are conjugate acid–base pairs in water.

Acids and Bases

Acid and Base Strength

- *In every acid–base reaction, equilibrium favors transfer of the proton from the stronger acid to the stronger base to form the weaker acid and the weaker base.*
- **The stronger an acid, the weaker its conjugate base; the stronger a base, the weaker its conjugate acid.**

➢ $HCl(aq) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$

➢ H_2O is a much stronger base than Cl^-, so the equilibrium lies far to the right ($K >> 1$).

❖ $CH_3COOH(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + CH_3COO^-(aq)$

❖ Acetate is a stronger base than H_2O, so the equilibrium favors the left side ($K < 1$).

SAMPLE EXERCISE 16.3 Predicting the Position of a Proton-Transfer Equilibrium

For the following proton-transfer reaction use Figure 16.4 to predict whether the equilibrium lies to the left ($K_c < 1$) or to the right ($K_c > 1$):

$$HSO_4^-(aq) + CO_3^{2-}(aq) \rightleftharpoons SO_4^{2-}(aq) + HCO_3^-(aq)$$

SAMPLE EXERCISE 16.3 Predicting the Position of a Proton-Transfer Equilibrium

For the following proton-transfer reaction use Figure 16.4 to predict whether the equilibrium lies to the left ($K_c < 1$) or to the right ($K_c > 1$):

$$HSO_4^-(aq) + CO_3^{2-}(aq) \rightleftharpoons SO_4^{2-}(aq) + HCO_3^-(aq)$$

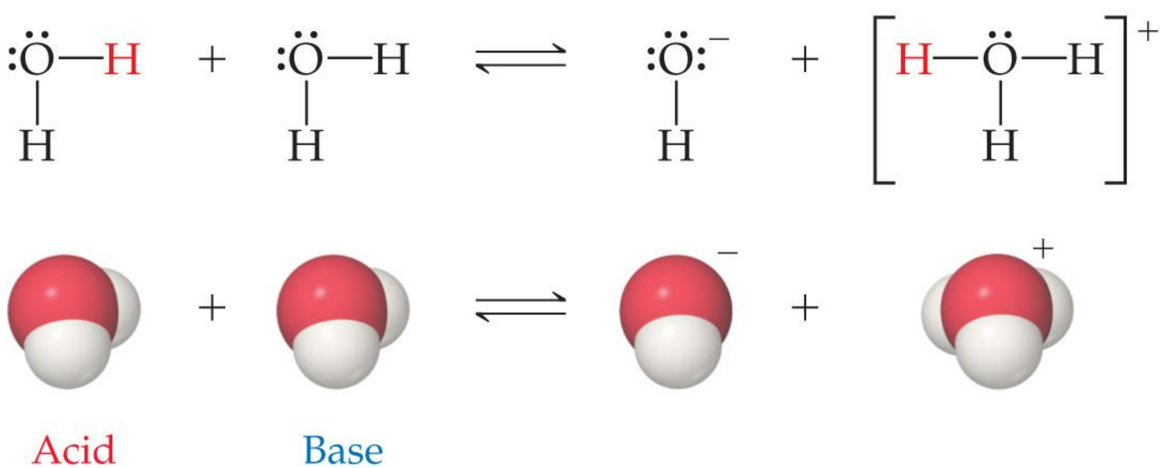
Solve The CO_3^{2-} ion appears lower in the right-hand column in Figure 16.4 and is therefore a stronger base than SO_4^{2-}. Therefore, CO_3^{2-} will get the proton preferentially to become HCO_3^-, while SO_4^{2-} will remain mostly unprotonated. The resulting equilibrium lies to the right, favoring products (that is, $K_c > 1$):

$$\underset{\text{Acid}}{HSO_4^-(aq)} + \underset{\text{Base}}{CO_3^{2-}(aq)} \rightleftharpoons \underset{\text{Conjugate base}}{SO_4^{2-}(aq)} + \underset{\text{Conjugate acid}}{HCO_3^-(aq)} \quad K_c > 1$$

Autoionization of Water

- Water is **amphoteric**.
- In pure water, a few molecules act as bases and a few act as acids.
- This is referred to as **autoionization**.

$$H_2O(l) + H_2O(l) \rightleftharpoons OH^-(aq) + H_3O^+(aq)$$



Ion Product Constant

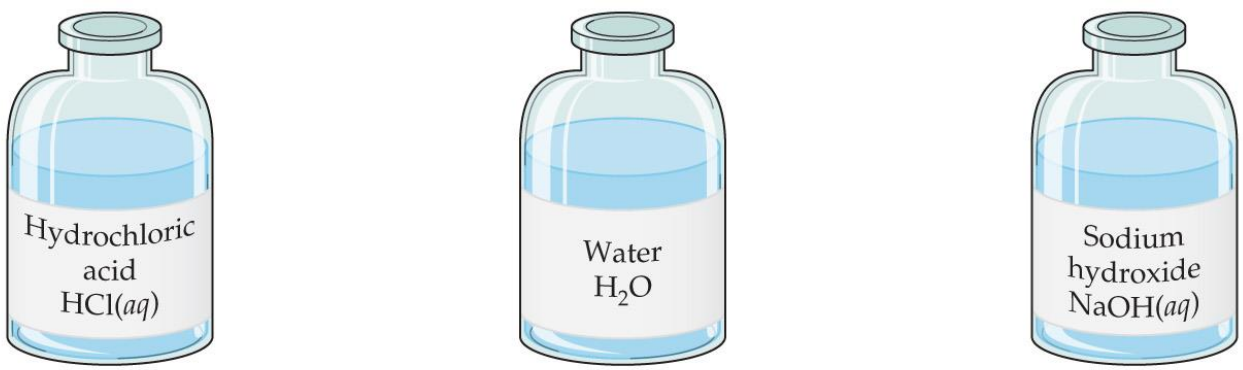
- The equilibrium expression for this process is

$$K_c = [H_3O^+][OH^-]$$

- This special equilibrium constant is referred to as the **ion product constant** for water, K_w.
- At 25 ° C, $K_w = 1.0 \times 10^{-14}$

Aqueous Solutions Can Be Acidic, Basic, or Neutral

- If a solution is neutral, $[H^+] = [OH^-]$.
- If a solution is acidic, $[H^+] > [OH^-]$.
- If a solution is basic, $[H^+] < [OH^-]$.



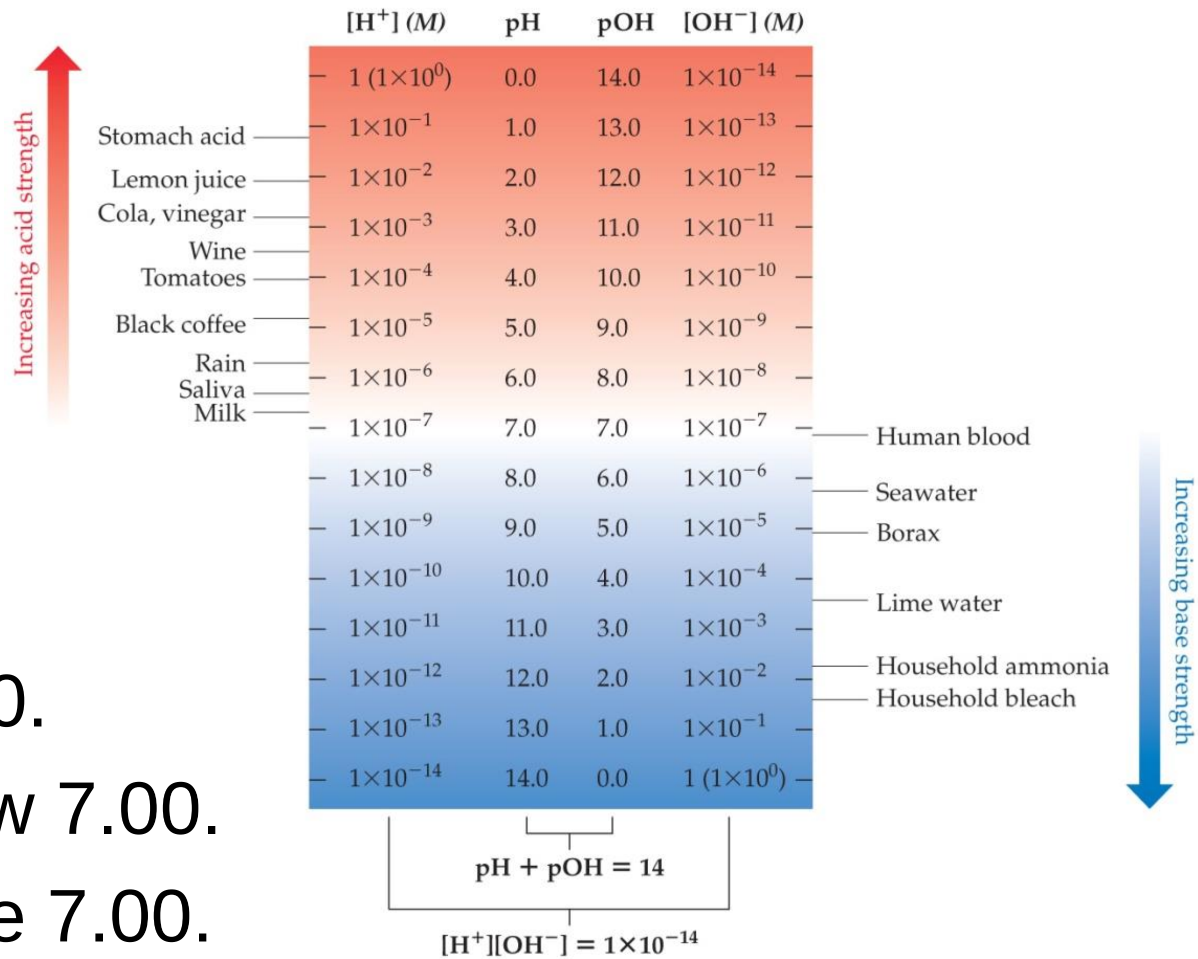
Acidic solution
$[H^+] > [OH^-]$
$[H^+][OH^-] = 1.0 \times 10^{-14}$

Neutral solution
$[H^+] = [OH^-]$
$[H^+][OH^-] = 1.0 \times 10^{-14}$

Basic solution
$[H^+] < [OH^-]$
$[H^+][OH^-] = 1.0 \times 10^{-14}$

pH

- pH is a method of reporting hydronium ion concentration.
- pH = –log[H$^+$]
- Neutral pH is 7.00.
- Acidic pH is below 7.00.
- Basic pH is above 7.00.



Acids and Bases

SAMPLE EXERCISE 16.5 Calculating $[H^+]$ from $[OH^-]$

Calculate the concentration of $H^+(aq)$ in **(a)** a solution in which $[OH^-]$ is 0.010 *M*, **(b)** a solution in which $[OH^-]$ is $1.8 \times 10^{-9} M$. *Note:* In this problem and all that follow, we assume, unless stated otherwise, that the temperature is 25 °C.

SAMPLE EXERCISE 16.5 Calculating $[H^+]$ from $[OH^-]$

Calculate the concentration of $H^+(aq)$ in **(a)** a solution in which $[OH^-]$ is 0.010 *M*, **(b)** a solution in which $[OH^-]$ is $1.8 \times 10^{-9} M$. *Note:* In this problem and all that follow, we assume, unless stated otherwise, that the temperature is 25 °C.

Solve

(a) Using Equation 16.16, we have

$$[H^+][OH^-] = 1.0 \times 10^{-14}$$

$$[H^+] = \frac{(1.0 \times 10^{-14})}{[OH^-]} = \frac{1.0 \times 10^{-14}}{0.010} = 1.0 \times 10^{-12}\,M$$

This solution is basic because

$$[OH^-] > [H^+]$$

(b) In this instance

$$[H^+] = \frac{(1.0 \times 10^{-14})}{[OH^-]} = \frac{1.0 \times 10^{-14}}{1.8 \times 10^{-9}} = 5.6 \times 10^{-6}\,M$$

This solution is acidic because

$$[H^+] > [OH^-]$$

Acids and Bases

Other "p" Scales

- The "p" in pH tells us to take the –log of a quantity (in this case, hydronium ions).

- Some other "p" systems are
  - pOH: $-\log[OH^-]$
  - pK_w: $-\log K_w$

Relating pH and pOH

Because

$$[H_3O^+][OH^-] = K_w = 1.0 \times 10^{-14}$$

we can take the –log of the equation

$$-\log[H_3O^+] + -\log[OH^-] = -\log K_w = 14.00$$

which results in

$$pH + pOH = pK_w = 14.00$$

SAMPLE EXERCISE 16.7 Calculating $[H^+]$ from pOH

A sample of freshly pressed apple juice has a pOH of 10.24. Calculate $[H^+]$.

SAMPLE EXERCISE 16.7 Calculating $[H^+]$ from pOH

A sample of freshly pressed apple juice has a pOH of 10.24. Calculate $[H^+]$.

Solve From Equation 16.20, we have

$$pH = 14.00 - pOH$$

$$pH = 14.00 - 10.24 = 3.76$$

Next we use Equation 16.17:

$$pH = -\log[H^+] = 3.76$$

Thus,

$$\log[H^+] = -3.76$$

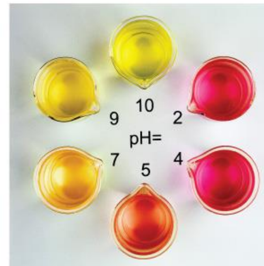
To find $[H^+]$, we need to determine the *antilogarithm* of -3.76. Your calculator will show this command as 10^x or INV log (these functions are usually above the log key). We use this function to perform the calculation:

$$[H^+] = \text{antilog}(-3.76) = 10^{-3.76} = 1.7 \times 10^{-4}\,M$$

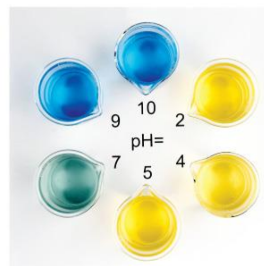
How Do We Measure pH?

- **<u>Indicators</u>**, including litmus paper, are used for less accurate measurements; an indicator is one color in its acid form and another color in its basic form.

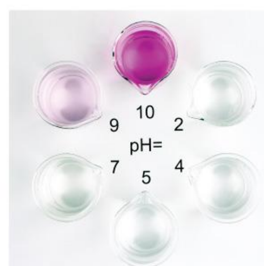
- **<u>pH meters</u>** are used for accurate measurement of pH; electrodes indicate small changes in voltage to detect pH.



Methyl red



Bromthymol blue



Phenolphthalein

Strong Acids

- You will recall that the seven strong acids are HCl, HBr, HI, HNO_3, H_2SO_4, $HClO_3$, and $HClO_4$.
- These are, by definition, strong electrolytes and exist totally as ions in aqueous solution; e.g.,

$$HA + H_2O \rightarrow H_3O^+ + A^-$$

- So, for the monoprotic strong acids,

$$[H_3O^+] = [\text{acid}]$$

SAMPLE EXERCISE 16.8 Calculating the pH of a Strong Acid

What is the pH of a 0.040 *M* solution of $HClO_4$?

SAMPLE EXERCISE 16.8 Calculating the pH of a Strong Acid

What is the pH of a 0.040 *M* solution of $HClO_4$?

Solve

$$\text{pH} = -\log(0.040) = 1.40$$

Check Because $[H^+]$ lies between 1×10^{-2} and 1×10^{-1}, the pH will be between 2.0 and 1.0. Our calculated pH falls within the estimated range. Furthermore, because the concentration has two significant figures, the pH has two decimal places.

Strong Bases

- Strong bases are the soluble hydroxides, which are the alkali metal and heavier alkaline earth metal hydroxides (Ca^{2+}, Sr^{2+}, and Ba^{2+}).

- Again, these substances dissociate completely in aqueous solution; e.g.,

$$MOH(aq) \rightarrow M^{+}(aq) + OH^{-}(aq) \quad \textit{or}$$

$$M(OH)_2(aq) \rightarrow M^{2+}(aq) + 2\ OH^{-}(aq)$$

SAMPLE EXERCISE 16.9 Calculating the pH of a Strong Base

What is the pH of **(a)** a 0.028 *M* solution of NaOH, **(b)** a 0.0011 *M* solution of $Ca(OH)_2$?

Acids and Bases

SAMPLE EXERCISE 16.9 Calculating the pH of a Strong Base

What is the pH of **(a)** a 0.028 M solution of NaOH, **(b)** a 0.0011 M solution of $Ca(OH)_2$?

Solve

(a) NaOH dissociates in water to give one OH^- ion per formula unit. Therefore, the OH^- concentration for the solution in (a) equals the stated concentration of NaOH, namely 0.028 M.

Method 1:

$$[H^+] = \frac{1.0 \times 10^{-14}}{0.028} = 3.57 \times 10^{-13}\,M \qquad pH = -\log(3.57 \times 10^{-13}) = 12.45$$

Method 2:

$$pOH = -\log(0.028) = 1.55 \qquad pH = 14.00 - pOH = 12.45$$

(b) $Ca(OH)_2$ is a strong base that dissociates in water to give *two* OH^- ions per formula unit. Thus, the concentration of $OH^-(aq)$ for the solution in part (b) is $2 \times (0.0011\,M) = 0.0022\,M$.

Method 1:

$$[H^+] = \frac{1.0 \times 10^{-14}}{0.0022} = 4.55 \times 10^{-12}\,M \qquad pH = -\log(4.55 \times 10^{-12}) = 11.34$$

Method 2:

$$pOH = -\log(0.0022) = 2.66 \qquad pH = 14.00 - pOH = 11.34$$

Weak Acids

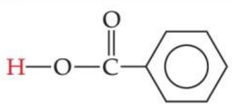
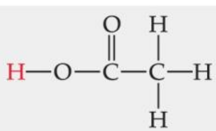
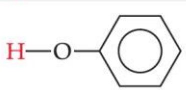
- For a weak acid, the equation for its dissociation is

$$HA(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + A^-(aq)$$

- Since it is an equilibrium, there is an equilibrium constant related to it, called the **acid-dissociation constant**, K_a:

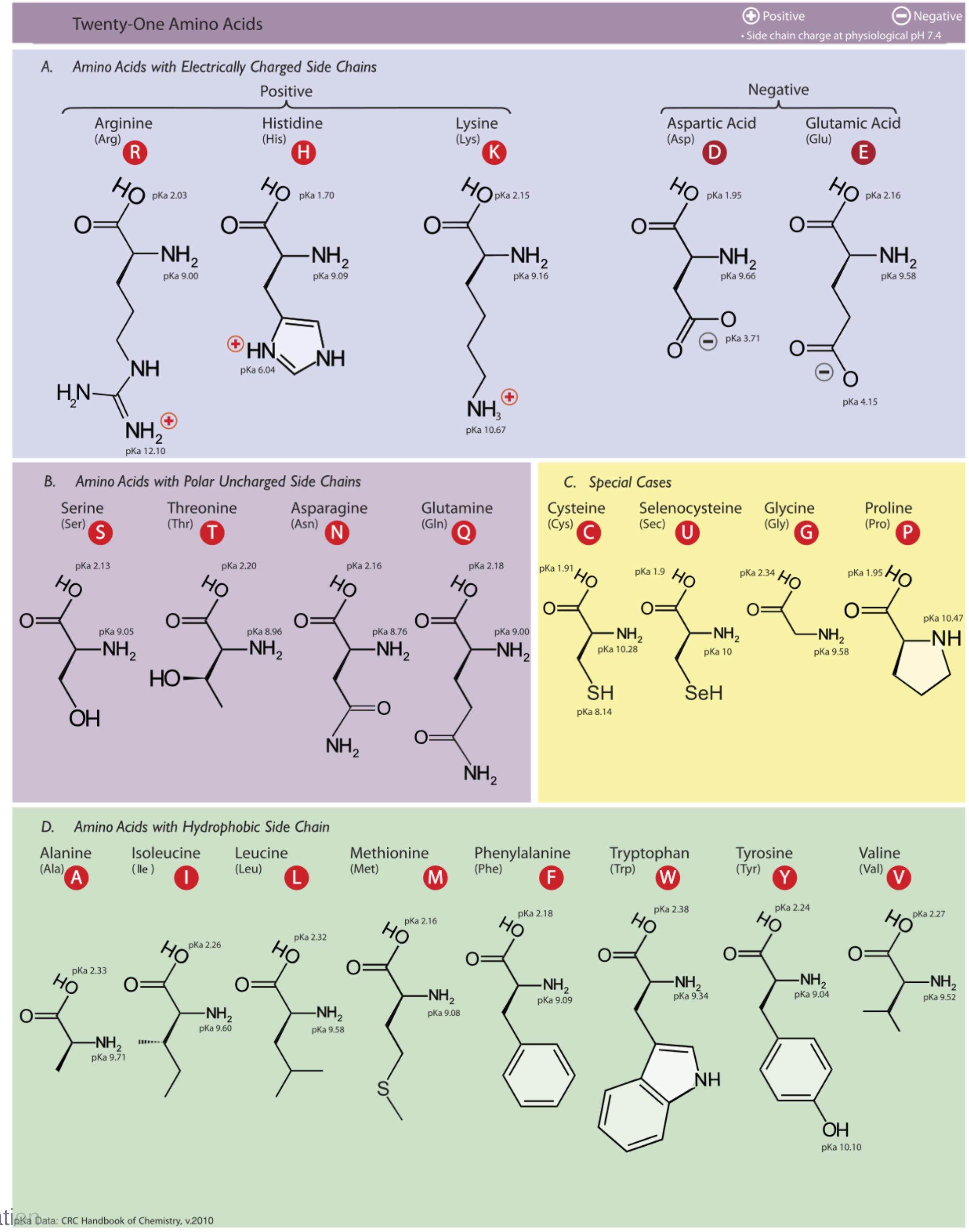
$$K_a = [H_3O^+][A^-] / [HA]$$

Table 16.2 Some Weak Acids in Water at 25 °C

Acid	Structural Formula*	Conjugate Base	K_a
Chlorous ($HClO_2$)	H—O—Cl—O	ClO_2^-	1.0×10^{-2}
Hydrofluoric (HF)	H—F	F^-	6.8×10^{-4}
Nitrous (HNO_2)	H—O—N=O	NO_2^-	4.5×10^{-4}
Benzoic (C_6H_5COOH)		$C_6H_5COO^-$	6.3×10^{-5}
Acetic (CH_3COOH)		CH_3COO^-	1.8×10^{-5}
Hypochlorous (HOCl)	H—O—Cl	OCl^-	3.0×10^{-8}
Hydrocyanic (HCN)	H—C≡N	CN^-	4.9×10^{-10}
Phenol (HOC_6H_5)		$C_6H_5O^-$	1.3×10^{-10}

*The proton that ionizes is shown in red.

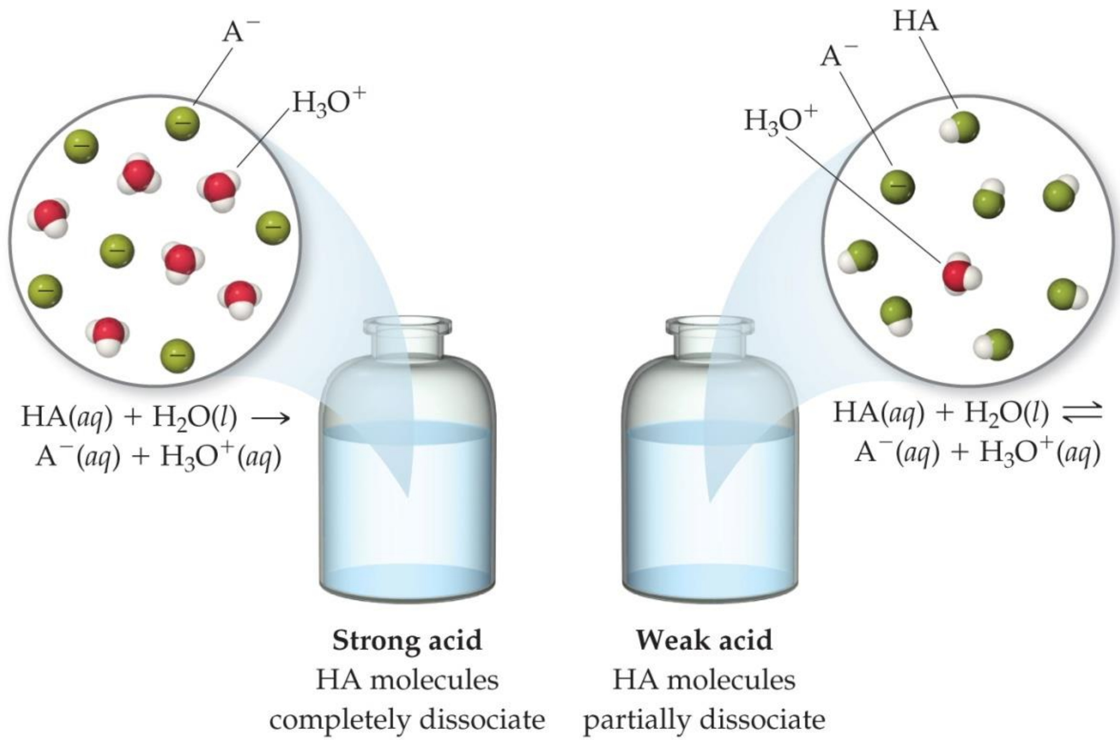
- The greater the value of K_a, the stronger is the acid.



Acids and Bases

Comparing Strong and Weak Acids

- What is present in solution for a strong acid versus a weak acid?
- Strong acids *completely* dissociate to ions.
- Weak acids only *partially* dissociate to ions.



Acids and Bases

Calculating K_a from the pH

- The pH of a 0.10 *M* solution of formic acid, HCOOH, at 25 °C is 2.38. Calculate K_a for formic acid at this temperature.

➢ We know that

$$K_a = \frac{[H_3O^+][HCOO^-]}{[HCOOH]}$$

➢ To calculate K_a, we need the equilibrium concentrations of all three things.

➢ We can find $[H_3O^+]$, which is the same as $[HCOO^-]$, from the pH.

➢ $[H_3O^+] = [HCOO^-] = 10^{-2.38} = 4.2 \times 10^{-3}$

Calculating K_a from pH

Now we can set up a table for equilibrium concentrations. We know initial HCOOH (0.10 *M*) and ion concentrations (0 *M*); we found equilibrium ion concentrations (4.2×10^{-3} *M*); so we calculate the change, then the equilibrium HCOOH concentration.

	[HCOOH], *M*	$[H_3O^+]$, *M*	$[HCOO^-]$, *M*
Initially	0.10	0	0
Change	-4.2×10^{-3}	$+4.2 \times 10^{-3}$	$+4.2 \times 10^{-3}$
Equilibrium	$0.10 - 4.2 \times 10^{-3} = 0.0958 = 0.10$	4.2×10^{-3}	4.2×10^{-3}

Calculating K_a from pH

- This allows us to calculate K_a by putting in the equilibrium concentrations.

$$K_a = \frac{[4.2 \times 10^{-3}][4.2 \times 10^{-3}]}{[0.10]}$$

$$= 1.8 \times 10^{-4}$$

Calculating Percent Ionization

- Percent ionization = $\dfrac{[H_3O^+]_{eq}}{[HA]_{initial}} \times 100$
- In this example,

$$[H_3O^+]_{eq} = 4.2 \times 10^{-3}\ M$$

$$[HCOOH]_{initial} = 0.10\ M$$

$$\text{Percent ionization} = \frac{4.2 \times 10^{-3}}{0.10} \times 100$$

$$= 4.2\%$$

Acids and Bases

Steps to Follow to Calculate pH Using K_a

1) Write the chemical equation for the ionization equilibrium.
2) Write the equilibrium constant expression.
3) Set up a table for Initial/Change in/Equilibrium Concentration to determine equilibrium concentrations as a function of change (x).
4) Substitute equilibrium concentrations into the equilibrium constant expression and solve for x. (Make assumptions if you can!)

- Calculate the pH of a 0.30 *M* solution of acetic acid, CH_3COOH, at 25 °C. $K_a = 1.8 \times 10^{-5}$

- Calculate the pH of a 0.30 *M* solution of acetic acid, CH_3COOH, at 25 °C. $K_a = 1.8 \times 10^{-5}$

1) $CH_3COOH + H_2O \rightleftharpoons H_3O^+ + CH_3COO^-$
2) $K_a = [H_3O^+][CH_3COO^-] / [CH_3COOH]$
3)

	CH_3COOH (*M*)	H_3O^+ (*M*)	CH_3COO^- (*M*)
Initial Concentration (*M*)	0.30	0	0
Change in Concentration (*M*)	$-x$	$+x$	$+x$
Equilibrium Concentration (*M*)	$0.30 - x$	x	x

4) $K_a = [H_3O^+][CH_3COO^-] / [CH_3COOH]$

$= (x)(x) / (0.30 - x)$

If we assume that $x \ll 0.30$ **(if smaller than 5% or 1/20)**, then $(0.30 - x) \approx 0.30$. The problem becomes easier, since we don't have to use the quadratic formula to solve it.

$K_a = 1.8 \times 10^{-5} = x^2 / 0.30$

so $x = 2.3 \times 10^{-3}$ (indeed, $x \ll 0.30$)

$x = [H_3O^+]$, so $pH = -\log(2.3 \times 10^{-3}) = 2.64$

Acids and Bases

SAMPLE EXERCISE 16.12 Using K_a to Calculate pH

Calculate the pH of a 0.20 M solution of HCN. (Refer to Table 16.2 or Appendix D for the value of K_a.)

K_a for HCN is 4.9×10^{-10}.

Acids and Bases

SAMPLE EXERCISE 16.12 Using K_a to Calculate pH

Calculate the pH of a 0.20 *M* solution of HCN. (Refer to Table 16.2 or Appendix D for the value of K_a.)

K_a for HCN is 4.9×10^{-10}.

Solve Writing both the chemical equation for the ionization reaction that forms $H^+(aq)$ and the equilibrium-constant (K_a) expression for the reaction:

$$HCN(aq) \rightleftharpoons H^+(aq) + CN^-(aq)$$

$$K_a = \frac{[H^+][CN^-]}{[HCN]} = 4.9 \times 10^{-10}$$

Next, we tabulate the concentrations of the species involved in the equilibrium reaction, letting $x = [H^+]$ at equilibrium:

	$HCN(aq)$ $\rightleftharpoons$	$H^+(aq)$ +	$CN^-(aq)$
Initial concentration (*M*)	0.20	0	0
Change in concentration (*M*)	$-x$	$+x$	$+x$
Equilibrium concentration (*M*)	$(0.20 - x)$	x	x

Substituting the equilibrium concentrations into the equilibrium-constant expression yields

$$K_a = \frac{(x)(x)}{0.20 - x} = 4.9 \times 10^{-10}$$

We next make the simplifying approximation that x, the amount of acid that dissociates, is small compared with the initial concentration of acid, $0.20 - x \simeq 0.20$. Thus,

$$\frac{x^2}{0.20} = 4.9 \times 10^{-10}$$

Solving for x, we have

$$x^2 = (0.20)(4.9 \times 10^{-10}) = 0.98 \times 10^{-10}$$

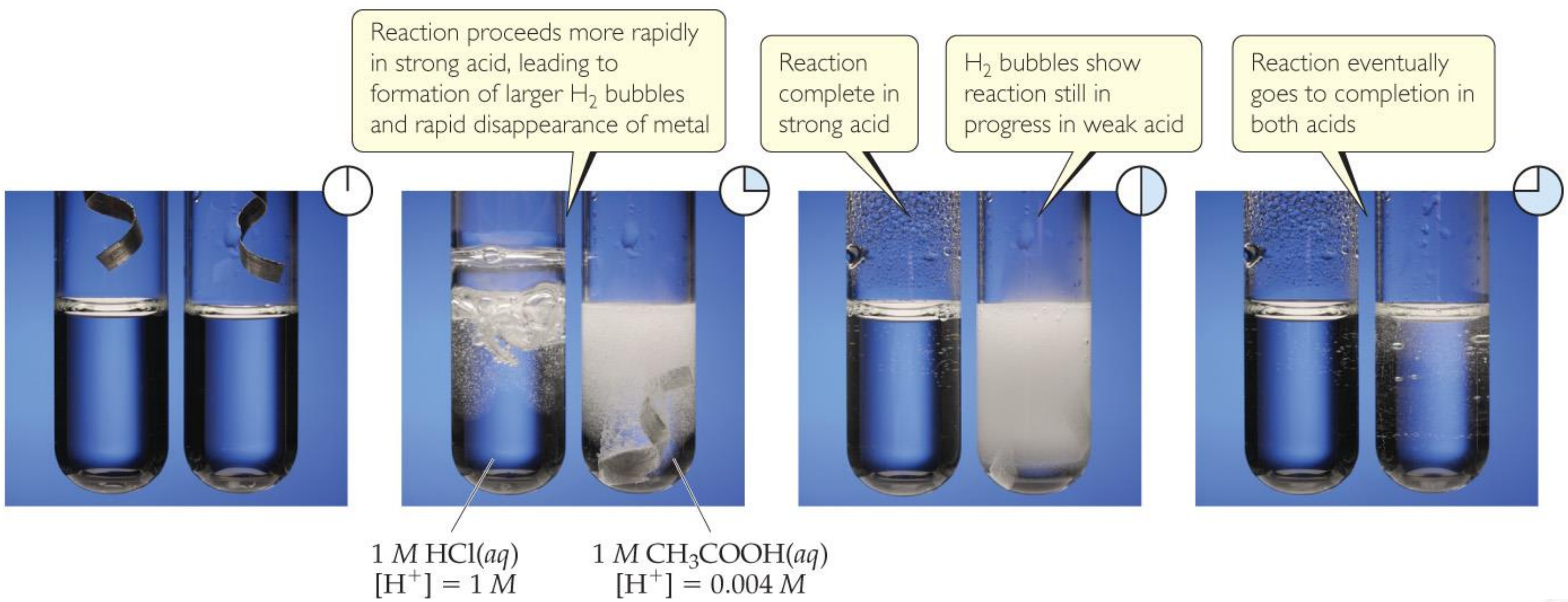
$$x = \sqrt{0.98 \times 10^{-10}} = 9.9 \times 10^{-6}\,M = [H^+]$$

A concentration of 9.9×10^{-6} *M* is much smaller than 5% of 0.20, the initial HCN concentration. Our simplifying approximation is therefore appropriate. We now calculate the pH of the solution:

$$pH = -\log[H^+] = -\log(9.9 \times 10^{-6}) = 5.00$$

Strong vs. Weak Acids—Another Comparison

- Strong Acid: $[H^+]_{eq} = [HA]_{init}$
- Weak Acid: $[H^+]_{eq} < [HA]_{init}$
- This creates a difference in conductivity and in rates of chemical reactions.



Polyprotic Acids

- Polyprotic acids have more than one acidic proton.
- It is always easier to remove the first proton than any successive proton.
- If the factor in the K_a values for the first and second dissociation has a difference of 3 or greater, the pH generally depends *only* on the first dissociation.

Table 16.3 Acid-Dissociation Constants of Some Common Polyprotic Acids

Name	Formula	K_{a1}	K_{a2}	K_{a3}
Ascorbic	$H_2C_6H_6O_6$	8.0×10^{-5}	1.6×10^{-12}	
Carbonic	H_2CO_3	4.3×10^{-7}	5.6×10^{-11}	
Citric	$H_3C_6H_5O_7$	7.4×10^{-4}	1.7×10^{-5}	4.0×10^{-7}
Oxalic	HOOC—COOH	5.9×10^{-2}	6.4×10^{-5}	
Phosphoric	H_3PO_4	7.5×10^{-3}	6.2×10^{-8}	4.2×10^{-13}
Sulfurous	H_2SO_3	1.7×10^{-2}	6.4×10^{-8}	
Sulfuric	H_2SO_4	Large	1.2×10^{-2}	
Tartaric	$C_2H_2O_2(COOH)_2$	1.0×10^{-3}	4.6×10^{-5}	

SAMPLE EXERCISE 16.14 Calculating the pH of a Solution of a Polyprotic Acid

The solubility of CO_2 in water at 25 °C and 0.1 atm is 0.0037 *M*. The common practice is to assume that all the dissolved CO_2 is in the form of carbonic acid (H_2CO_3), which is produced in the reaction

$$CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$

What is the pH of a 0.0037 *M* solution of H_2CO_3?

SAMPLE EXERCISE 16.14 Calculating the pH of a Solution of a Polyprotic Acid

The solubility of CO_2 in water at 25 °C and 0.1 atm is 0.0037 *M*. The common practice is to assume that all the dissolved CO_2 is in the form of carbonic acid (H_2CO_3), which is produced in the reaction

$$CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$$

What is the pH of a 0.0037 *M* solution of H_2CO_3?

Solve Proceeding as in Sample Exercises 16.12 and 16.13, we can write the equilibrium reaction and equilibrium concentrations as

$$H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$$

	$H_2CO_3(aq)$	$H^+(aq)$	$HCO_3^-(aq)$
Initial concentration (*M*)	0.0037	0	0
Change in concentration (*M*)	$-x$	$+x$	$+x$
Equilibrium concentration (*M*)	$(0.0037 - x)$	x	x

The equilibrium-constant expression is

$$K_{a1} = \frac{[H^+][HCO_3^-]}{[H_2CO_3]} = \frac{(x)(x)}{0.0037 - x} = 4.3 \times 10^{-7}$$

Solving this quadratic equation, we get

$$x = 4.0 \times 10^{-5}\ M$$

Alternatively, because K_{a1} is small, we can make the simplifying approximation that x is small, so that

$$0.0037 - x \simeq 0.0037$$

Thus,

$$\frac{(x)(x)}{0.0037} = 4.3 \times 10^{-7}$$

Solving for x, we have

$$x^2 = (0.0037)(4.3 \times 10^{-7}) = 1.6 \times 10^{-9}$$

$$x = [H^+] = [HCO_3^-] = \sqrt{1.6 \times 10^{-9}} = 4.0 \times 10^{-5} M$$

Because we get the same value (to 2 significant figures) our simplifying assumption was justified. The pH is therefore

$$pH = -\log[H^+] = -\log(4.0 \times 10^{-5}) = 4.40$$

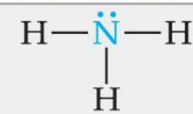
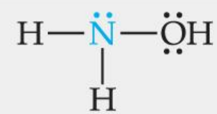
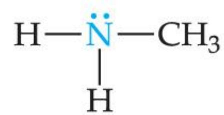
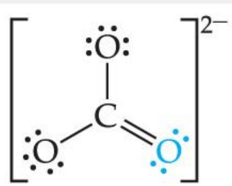
Weak Bases

- Ammonia, $NH_{3,}$ is a weak base.
- Like weak acids, weak bases have an equilibrium constant called the **base dissociation constant**.
- Equilibrium calculations work the *same* as for acids, using the base dissociation constant instead.

$$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq) \quad K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

Base Dissociation Constants

Table 16.4 Some Weak Bases in Water at 25 °C

Base	Structural Formula*	Conjugate Acid	K_b
Ammonia (NH_3)		NH_4^+	1.8×10^{-5}
Pyridine (C_5H_5N)		$C_5H_5NH^+$	1.7×10^{-9}
Hydroxylamine ($HONH_2$)		$HONH_3^+$	1.1×10^{-8}
Methylamine (CH_3NH_2)		$CH_3NH_3^+$	4.4×10^{-4}
Hydrosulfide ion (HS^-)		H_2S	1.8×10^{-7}
Carbonate ion (CO_3^{2-})		HCO_3^-	1.8×10^{-4}
Hypochlorite ion (ClO^-)		HClO	3.3×10^{-7}

*The atom that accepts the proton is shown in blue.

- What is the pH of 0.15 *M* NH_3?

- What is the pH of 0.15 *M* NH_3?

1) $NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$

2) $K_b = [NH_4^+][OH^-] / [NH_3] = 1.8 \times 10^{-5}$

3)

	NH_3 (*M*)	NH_4^+ (*M*)	OH^- (*M*)
Initial Concentration (*M*)	0.15	0	0
Change in Concentration (*M*)	$-x$	$+x$	$+x$
Equilibrium Concentration (*M*)	$0.15 - x$	x	x

Acids and Bases

4) $1.8 \times 10^{-5} = x^2 / (0.15 - x)$

If we assume that $x << 0.15$, $0.15 - x \approx 0.15$.

Then: $1.8 \times 10^{-5} = x^2 / 0.15$

and: $x = 1.6 \times 10^{-3}$

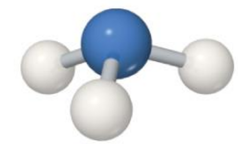
Note: x is the molarity of OH^-, so $-\log(x)$ will be the pOH (pOH = 2.80) and [14.00 – pOH] is pH (pH = 11.20).

Types of Weak Bases

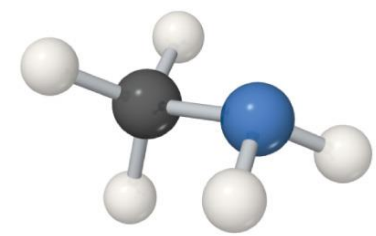
- Two main categories

1) Neutral substances with an atom that has a nonbonding pair of electrons that can accept H^+ (like ammonia and the **amines**)

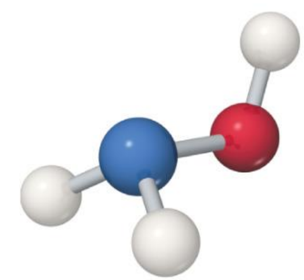
2) Anions of weak acids - conjugate base



Ammonia
NH_3



Methylamine
CH_3NH_2



Hydroxylamine
NH_2OH

Relationship between K_a and K_b

Table 16.5 Some Conjugate Acid–Base Pairs

Acid	K_a	Base	K_b
HNO_3	(Strong acid)	NO_3^-	(Negligible basicity)
HF	6.8×10^{-4}	F^-	1.5×10^{-11}
CH_3COOH	1.8×10^{-5}	CH_3COO^-	5.6×10^{-10}
H_2CO_3	4.3×10^{-7}	HCO_3^-	2.3×10^{-8}
NH_4^+	5.6×10^{-10}	NH_3	1.8×10^{-5}
HCO_3^-	5.6×10^{-11}	CO_3^{2-}	1.8×10^{-4}
OH^-	(Negligible acidity)	O^{2-}	(Strong base)

For a conjugate acid–base pair, K_a and K_b are related in this way:

$$K_a \text{ (an acid)} \times K_b \text{ (its conjugate base)} = K_w$$

Therefore, if you know one of them, you can calculate the other.

Acids and Bases

Acid–Base Properties of Salts

- Many ions react with water to create H^+ or OH^-. The reaction with water is often called **hydrolysis**.
- To determine whether a salt is an acid or a base, you need to look at the cation and anion separately.
- The **cation** can be acidic or neutral.
- The **anion** can be acidic, basic, or neutral.

Anions

- Anions of strong acids are neutral. For example, Cl^- will *not* react with water, so OH^- can't be formed.

- Anions of weak acids are conjugate bases, so they create OH^- in water; e.g., $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

- Protonated anions from polyprotic acids can be acids or bases: If $K_a > K_b$, the anion will be acidic; if $K_b > K_a$, the anion will be basic. **For example, HCO_3^- (**comparing its K_a and K_b**)**

Cations

- Group I or Group II (Ca^{2+}, Sr^{2+}, or Ba^{2+}) metal cations are neutral.
- Polyatomic cations are typically the conjugate acids of a weak base; e.g., NH_4^+.
- Transition and post-transition metal cations are acidic. Why? (There are no H atoms in these cations!) Because they are Lewis acids (e-pair acceptors).

Acids and Bases

Hydrated Cations

- Transition and post-transition metals form hydrated cations (coordination with water).
- The water attached to the metal is termed an aqua ligand and is more acidic than free water molecules, making the hydrated ions acidic.

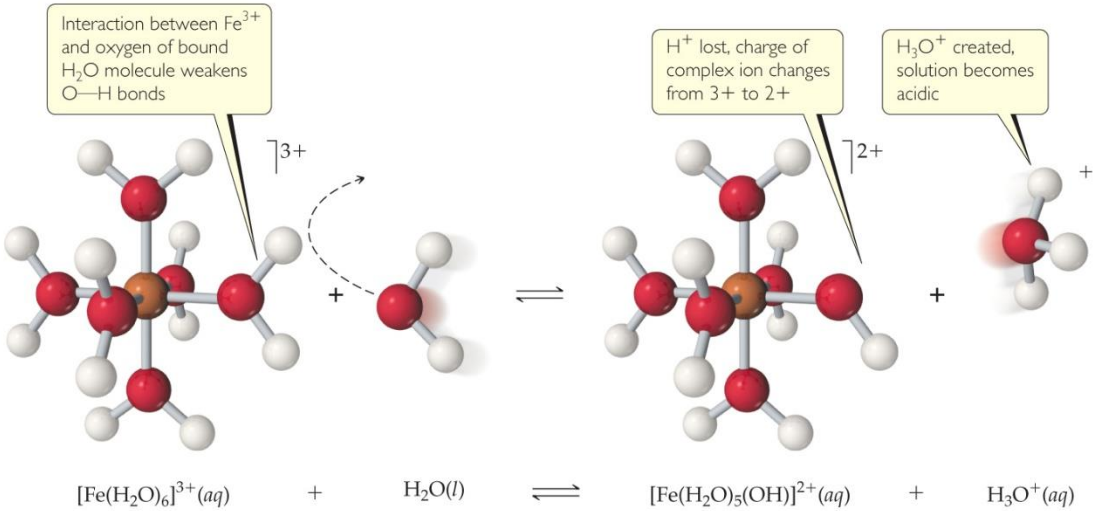


Table 16.6 Acid-Dissociation Constants for Metal Cations in Aqueous Solution at 25 °C

Cation	K_a
Fe^{2+}	3.2×10^{-10}
Zn^{2+}	2.5×10^{-10}
Ni^{2+}	2.5×10^{-11}
Fe^{3+}	6.3×10^{-3}
Cr^{3+}	1.6×10^{-4}
Al^{3+}	1.4×10^{-5}

Salt Solutions—Acidic, Basic, or Neutral?

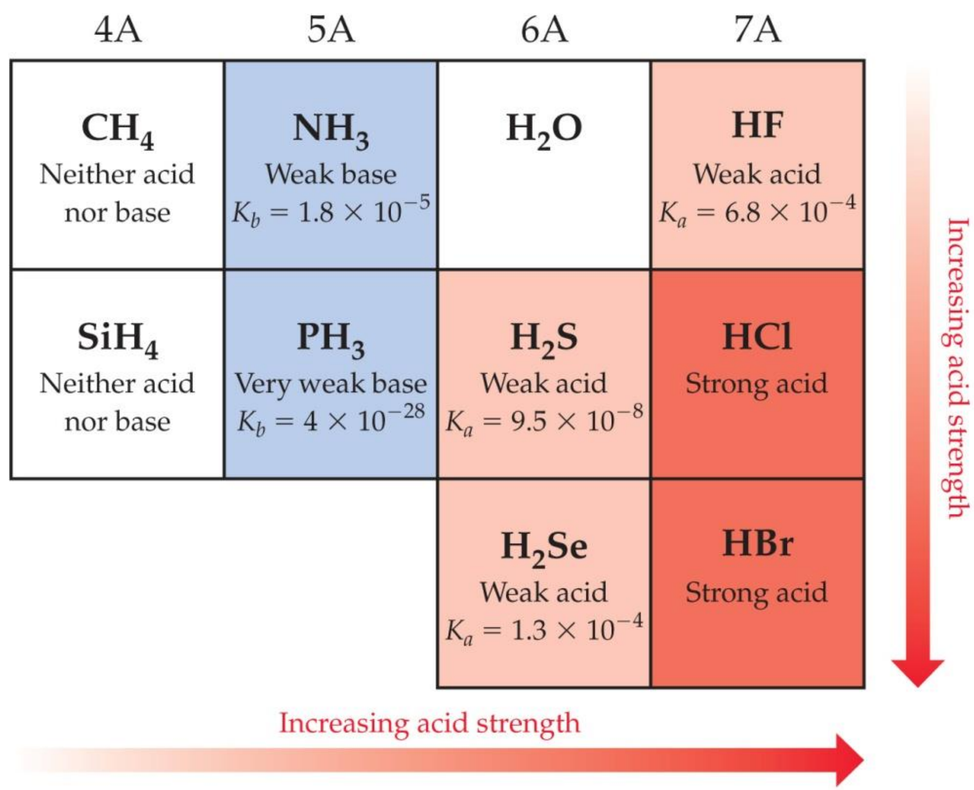
1) Group I/II metal cation (neutral) with anion of a strong acid: neutral
2) Group I/II metal cation (neutral) with anion of a weak acid: basic (like the anion)
3) Transition/Post-transition metal cation (Lewis acidic) or polyatomic cation with anion of a strong acid: acidic (like the cation)
4) Transition/Post-transition metal cation (Lewis acidic) or polyatomic cation with anion of a weak acid: compare K_a (of the cation) and K_b (of the anion); whichever is greater dictates what the salt is.

Factors that Affect Acid Strength

1) H—A bond must be polarized with δ+ on the H atom and δ– on the A atom
2) Bond strength: Weaker bonds can be broken more easily, making the acid stronger.
3) Stability of A^-: More stable anion means stronger acid.

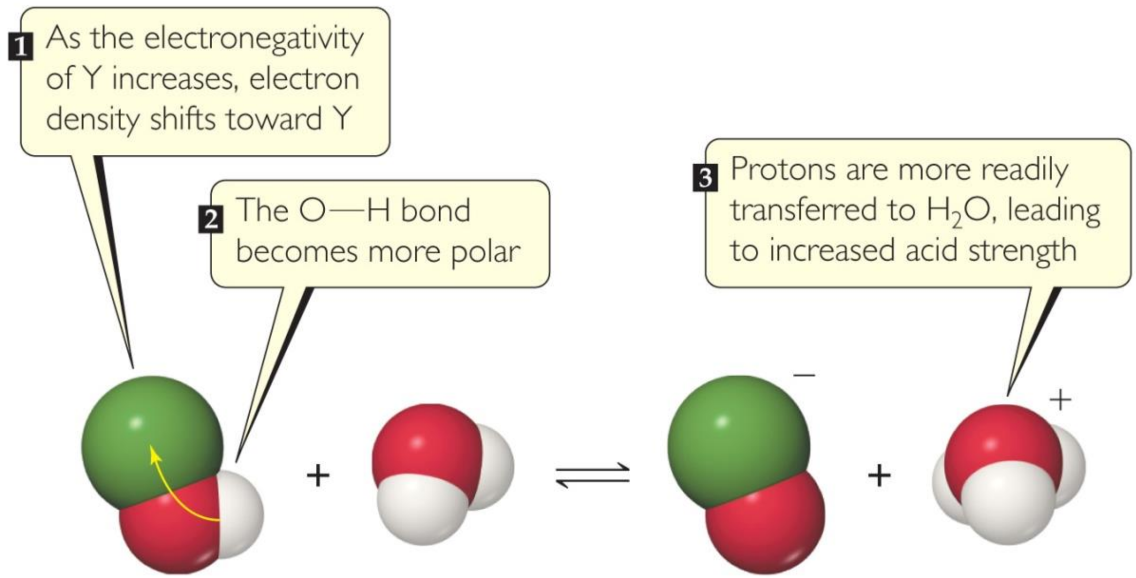
Binary Acids

- Binary acids consist of H and one other element.
- Within a group, H—A bond strength is generally the most important factor.
- Within a period, bond polarity is the most important factor to determine acid strength.



Oxyacids

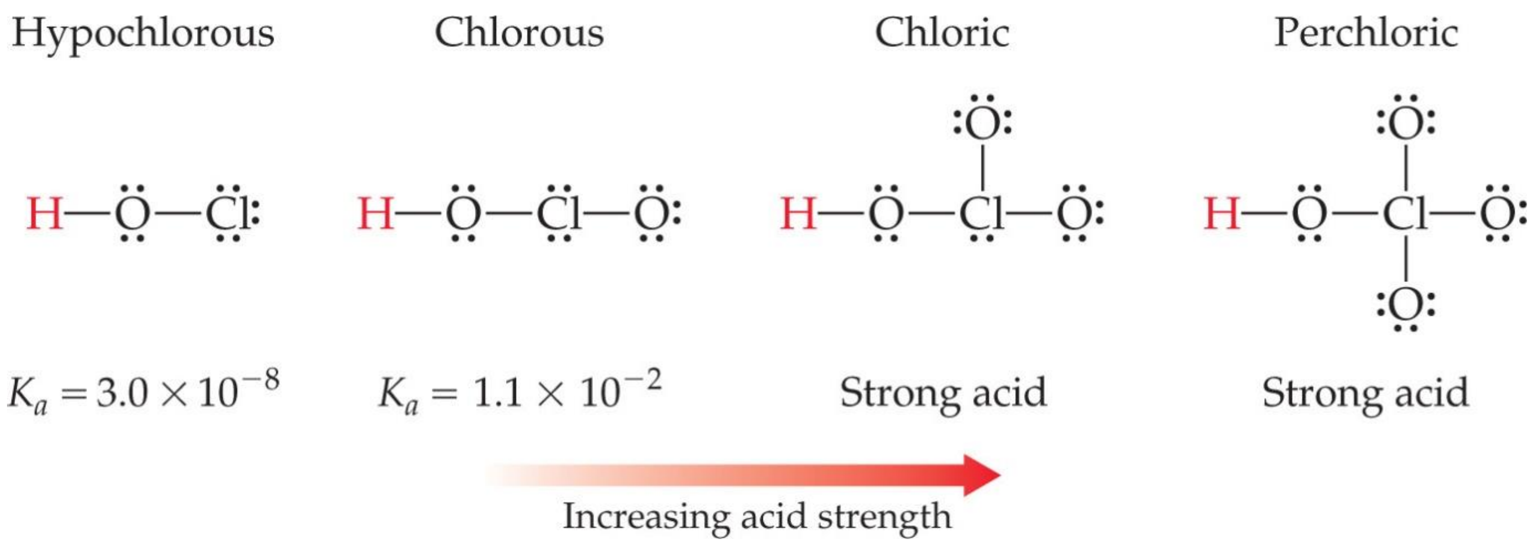
- **Oxyacids** consist of H, O, and one other element, which is a nonmetal Y.
- Generally, as the electronegativity of the nonmetal Y increases, the acidity increases for acids with the same structure.



Substance	Y—OH	Electronegativity of Y	Dissociation constant
Hypochlorous acid	Cl—OH	3.0	$K_a = 3.0 \times 10^{-8}$
Hypobromous acid	Br—OH	2.8	$K_a = 2.5 \times 10^{-9}$
Hypoiodous acid	I—OH	2.5	$K_a = 2.3 \times 10^{-11}$
Water	H—OH	2.1	$K_w = 1.0 \times 10^{-14}$

Oxyacids with Same "Other" Element

- If an element can form more than one oxyacid, the oxyacid with more O atoms (O is an atom with high electronegativity!) is more acidic; e.g., sulfuric acid versus sulfurous acid.
- Another way of saying it: If the oxidation number increases, the acidity increases.



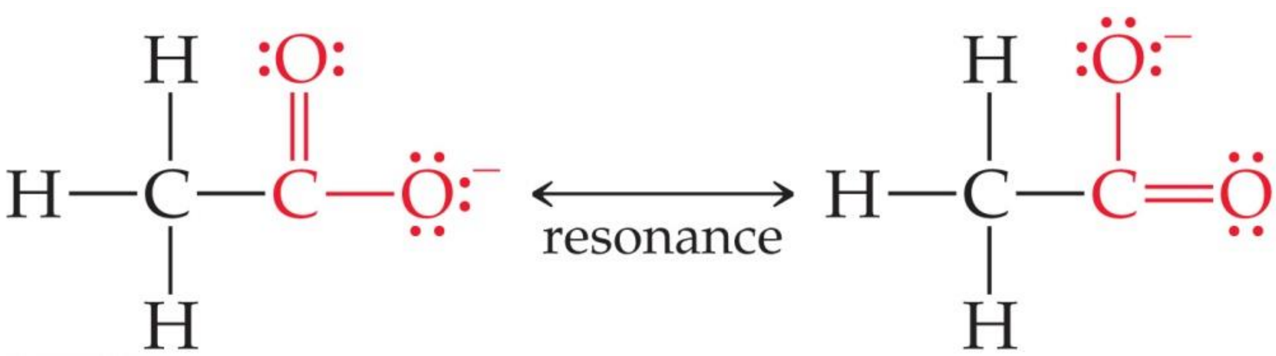
Carboxylic Acids

- **Carboxylic acids** are organic acids containing the —COOH group.

- Factors contributing to their acidic behavior:

➢ Other O attached to C draws electron density from O—H bond, increasing polarity.

➢ Its conjugate base (carboxylate anion) has resonance forms to stabilize the anion.



Acids and Bases